

Intelligent Association

ROUGH DRAFT

Spencer Randolph
 Rochester Institute of Technology
 Rochester, USA
 smr9985@rit.edu

Abstract—Radar and computer vision systems are commonly used to detect and track targets over time. In these systems, detections produced by a sensor are passed to a tracking algorithm, which associates new detections with existing tracks in order to maintain target identity across time. This association process also supports track management decisions, such as initiating new tracks, updating confirmed tracks, and terminating stale or invalid tracks. In noisy or cluttered environments, reliable association can be difficult due to measurement noise, missed detections, false alarms, and closely spaced targets. This project proposes a classical multi-object tracking pipeline that uses a learned model to estimate association scores between tracks and detection pairs instead of relying only on a standard distance based metric. The learned association model is evaluated against traditional distance based association using simulated radar style data with noisy measurements, false positives, and missed detections.

Index Terms—Tracking, Association, MLP, CV, AI/ML

I. INTRODUCTION

Tracking is the process of developing the best estimate of a target’s state over time based on noisy measurements [1]. More broadly, tracking provides a way to transform individual sensor observations into continuous target histories. Rather than treating each detection as an isolated measurement, a tracking system maintains information about targets across time, allowing it to estimate motion, preserve identity, and support real world decision making.

The value of tracking is clear in systems where decisions depend on how objects behave over time. A single detection may indicate that an object is present, but it provides limited information about the object’s motion, intent, or future position. A tracking system allows downstream processes to reason about movement patterns, anticipate future states, and respond to changes in the environment. This makes tracking an important component in applications such as autonomous navigation, robotics, air traffic monitoring, surveillance and defense systems.

Multi-object tracking extends this formulation from one target to a set of targets. The state of all objects at time t can be represented as $S_t = (s_t^1, s_t^2, \dots, s_t^{M_t})$, where M_t is the number of objects present in the scene at that time [2]. The corresponding observations are represented as O_t , and the goal is to estimate the sequence of states $S_{1:t}$ from the

sequence of observations $O_{1:t}$. This objective can be written as the maximum a posteriori estimation problem

$$S_{1:t}^* = \arg \max_{S_{1:t}} P(S_{1:t} | O_{1:t}). \quad (1)$$

In this view, multi-object tracking is the problem of recovering the most likely set of target predictions that explains the available observations.

Classical tracking systems often solve this problem using a predict associate update loop. As described in modern tracking system design, the tracker first predicts the next state of each existing track using a motion model [3]. These predictions are used to form distance based gates, which limit the observations considered as possible matches for each track, as shown in Fig. 1. The remaining candidate observations are then associated with tracks using an association rule or assignment method. Once associations are made, matched tracks are updated using the new measurements, while unmatched detections may initialize new tracks and unmatched tracks may be maintained or removed depending on the track management logic.

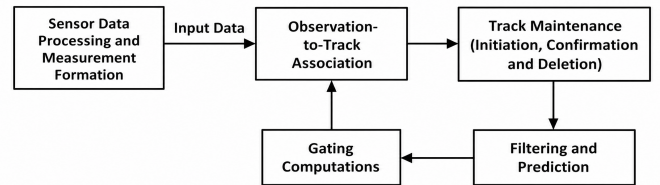


Fig. 1. Classical Tracking loop adapted from Blackman [3].

The association step becomes difficult because the tracker must choose among many possible assignments between predicted tracks and current observations. As the number of targets and detections increases, the number of possible assignments grows quickly, making data association a combinatorial problem [4]. This difficulty is increased in practical sensing environments where observations may include measurement noise, missed detections, clutter, and false alarms.

This project focuses on the data association step of multi-object tracking. Instead of using only a distance based association cost, a multilayer perceptron is trained to score predicted track and detection pairs. The learned score is then used within

a classical tracking pipeline and compared against a distance based baseline using simulated radar style data.

II. RELATED WORK

Data association has been studied through both classical probabilistic methods and modern learning based approaches. Classical methods attempt to manage uncertainty by reasoning over possible measurement to track assignments, while learning based methods attempt to estimate association scores or probabilities from data. This section reviews representative work in probabilistic association as well as machine learning approaches to data association.

One popular probabilistic method is joint probabilistic data association (JPDA) [5]. In this approach, association probabilities are computed between each track and the feasible observations within its gate. Rather than selecting only the nearest observation, the track is updated using a weighted combination of the possible matches, where each measurement contributes according to its association probability.

Another classical probabilistic association method is multiple hypothesis tracking (MHT) [6]. Unlike JPDA, which combines possible associations into a single weighted update, MHT maintains multiple possible association hypotheses over time and delays committing to a single explanation until more evidence is available.

More recent work has explored machine learning methods that learn association scores directly from data rather than relying only on hand designed probabilistic models or distance based rules. DeepDA is an LSTM learned association method designed for multi-target tracking in clutter [7]. Rather than replacing only a pairwise distance metric, DeepDA uses a recurrent network to estimate association probabilities from the current set of noisy radar measurements and existing tracks. In contrast, this project keeps the classical tracking pipeline intact and replaces only the hand-designed pairwise association cost with a multilayer perceptron score for individual track detection pairs.

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